

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | x_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | x_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F^2 = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.
- We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | x_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | x_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE_i + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

- Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.
- Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?
- The OLS fitted model from part (b), with usual and robust standard errors, is

$$\begin{array}{cccccc} \widehat{WAGE} = & -17.77 & + 2.50 EDUC & + 0.23 EXPER & + 3.23 METRO & - 4.20 FEMALE \\ (se) & (2.36) & (0.14) & (0.031) & (1.05) & (0.81) \\ (robse) & (2.50) & (0.16) & (0.029) & (0.84) & (0.80) \end{array}$$

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

- If we add $MARRIED$ to the model in part (b), we find that its t -value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

a.

$$H_0: \sigma_M^2 = \sigma_F^2$$

$$H_1: \sigma_M^2 \neq \sigma_F^2$$

$$\text{Males } N = 577$$

$$\hat{\sigma}_M^2 = \frac{SSE_M}{N-4} = \frac{97161.9174}{573} \approx 169.57$$

$$\text{Female } N = 1000 - 577 = 423$$

$$\hat{\sigma}_F^2 = 12.024 \quad \hat{\sigma}_F^2 = 144.58$$

$$F^* = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.57}{144.58} \approx 1.17$$

$$F_{0.05, 573, 419} \approx 1$$

$$\therefore 1.1728 > 1, \text{ We reject } H_0.$$

At 5% significant level, there's sufficient evidence to conclude that $\sigma_M^2 \neq \sigma_F^2$

$$b. H_0: \sigma_{single}^2 = \sigma_{married}^2$$

$$H_1: \sigma_{married}^2 > \sigma_{single}^2$$

$$N_S = 400$$

$$\hat{\sigma}_S^2 = \frac{SSE_S}{df_S} = \frac{56231.0382}{395} \approx 142.36$$

$$N_M = 600$$

$$\hat{\sigma}_M^2 = \frac{SSE_M}{df_M} = \frac{100703.0471}{595} \approx 169.25$$

$$F = \frac{\hat{\sigma}_{married}^2}{\hat{\sigma}_{single}^2} = \frac{169.25}{142.36} \approx 1.1889$$

$$F_{0.05, 595, 395} = 1.14$$

$$\therefore 1.1889 > 1.14 \therefore \text{We reject } H_0.$$

At 5% significant level, there's sufficient evidence to conclude that $\sigma_{married}^2 > \sigma_{single}^2$

$$c. NR^2 = 59.03$$

$$\chi_{0.05, 4}^2 = 9.488$$

$$\therefore 59.03 > 9.488 \therefore \text{We reject } H_0.$$

There's sufficient evidence to conclude that heteroskedasticity exists.

d. White Test

$$\text{test statistic} = 78.82$$

variables: EDUC, EXPER, METRO, FEMALE

square : EDUC², EXPER²

$$4 + 2 + C_2^4 = 12$$

$$\chi_{0.05, 12}^2 = 21.026$$

$\therefore 78.82 > 21.026 \therefore$ We reject H_0 .

There's sufficient evidence to conclude that heteroskedasticity exists.

e.

Narrower interval : EXPER (0.031 $\rightarrow$ 0.029) - METRO (1.05 $\rightarrow$ 0.84)

FEMALE (0.81 $\rightarrow$ 0.80)

Wider intervals : Intercept (2.36 $\rightarrow$ 2.50) - EDUC (0.14 $\rightarrow$ 0.16)

Under heteroskedasticity, the OLS standard errors are biased.

They can either overestimate or underestimate the true variance of the estimator.

The robust standard error simply corrects this bias, bringing the estimate closer to its true value. Therefore, it's perfectly mathematically consistent for some standard error to shrink (narrower intervals) while others increase (wider intervals) after the robust correction. So there's no inconsistency.

f.

They're compatible because they're testing two different statistical properties of wage distribution:

- The t-test in part (f) evaluates the effect of marriage on the conditional mean of wages.
- The F-Test in part (b) evaluates the effect of marriage on the conditional variance of wages.